RAWATAN KES *METHANOL/ETHYLENE GLYCOL POISONING*

DIAGNOSIS	• *History of methanol or ethylene glycol ingestion with a wide anion gap metabolic acidosis.* • *Patients may also present with visual disturbances, neurologic deficit, abdominal pain and vomiting, difficulty breathing or hyperventilation.* • *A **wide anion gap** can be used to diagnose poisoning in lieu of osmol gap if serum ethanol levels cannot be measured.* • *Anion gap measurement: [(K+Na)-(HCO3+Cl)], all units in mmol/l.* • *In early presentation a wide osmol gap will be detected but this gap normalizes later. Measured osmolality-calculated osmolality = Osmol gap.* • *Calculated osmolality: 2(Na + K) + Urea + Glucose + Ethanol (all units in mmol/l).* • *Normal osmol gap ranges from -2 ± 6 mOsm/L. A gap of more than 10 is clinically significant but is not diagnostic of toxic alcohol ingestion. A markedly elevated osmol gap is strongly suggestive of toxic alcohol ingestion.*
BASELINE INVESTIGATIONS	• *ABG* • *RP, RBS, blood ethanol level and serum osmolality level to calculate osmol gap.* • *Coagulation profile, LFT, CE, ECG as indicated.* • *CT Brain and CXR as indicated.* • *Blood and urine methanol levels can be measured in selected laboratories, the following must be sent:* i. *Blood: 3ml in grey cover bottle;* ii. *Urine: 25ml in universal container with sodium fluoride or 3 ml in grey cover bottle;* iii. *Specimen of alleged toxin;* iv. ***Sample should preferably be taken before commencing treatment.*** • *A serum formate level has been shown to accurately diagnose methanol poisoning but is currently unavailable in Malaysia.*

PRE-HOSPITAL CARE	• *Assess airway, breathing and circulation.* • *Provide ventilation and insert LMA if necessary.* • *Insert an IV line and administer fluid bolus if the patient is noted to be hypotensive.* • *Check DXT and administer Dextrose if DXT is less than 4.0 mmol.* • *Verify history of toxic alcohol ingestion and bring the bottle to ED if possible.* • *Triage to red or yellow zone after consultation with managing emergency physician.*
IN-HOSPITAL MANAGEMENT	• *Assess airway, breathing and circulation, stabilize as necessary. Fluid boluses to correct fluid deficit guided by urine output and preferably point of care ultrasound. Inotropes and vasopressors may be required.* • *Nasogastric aspiration with a protected airway can be performed in patients who present within 1 hour of consumption.* • *Activated charcoal is generally not recommended except in cases of significant, recent ingestion or in cases of co-ingestion of toxins that can be bound by activated charcoal.* • *Sodium bicarbonate as infusion can be given to correct severe acidosis after adequate fluid resuscitation.* • *The following vitamins should be administered:* *i. IV Folinic acid/Leucovorin calcium- 1mg/kg up to 50mg every 4-6 hours (Methanol);* *ii. IV Thiamine 100mg OD (Ethylene glycol);* *iii. IV/PO Pyridoxine 50mg every 4-6 hours (Ethylene glycol).* • *Fomepizole or ethanol should be administered as soon as possible to block conversion of toxic alcohols. Fomepizole is the preferred antidote as it is easily administered with fewer side effects and monitoring requirements as compared to ethanol. For ethanol to be an effective antidote serum levels must be carefully titrated and maintained.* • *Dialysis should be commenced immediately if patient shows signs of oliguric renal failure. Other indications are listed below.* • *Patients presenting in a comatose state or with status epilepticus will require a CT brain.* • *Treatment should be continued until blood methanol level is less than 20mg/dL and/or the patient shows clinical improvement with correction of metabolic acidosis, osmol gap, and renal function.*

Fomepizole :

- *A loading dose of 15mg/kg IV followed by 10mg/kg IV every 12 hours for the subsequent 2nd to 5th doses. Each dose should be diluted in 100ml NS or 5% Dextrose and infused over 30 minutes.*
- *Subsequent doses if required: 15mg/kg IV every 12 hours.*
- *In patients undergoing dialysis same dosages are administered except that the drug is given 6 hours after the first dose and 4 hours thereafter.*
- *Vital sign monitoring is required during infusion as the drug may cause bradycardia and hypotension.*
- *Due to its high cost, administration of fomepizole should be decided after consensus during breakouts. Fomepizole may be administered for the first day of treatment followed by ethanol infusion subsequently.*

Ethanol :

The chart below can be used as a guide for dosing:

LOADING DOSE	
Oral Ethanol 20%	***Intravenous Ethanol 10%***
0.8 gm/kg (5ml/kg) diluted in juice administered orally or via a nasogastric tube	*0.8gm/kg (10ml/kg) administered over 20 to 60 minutes as tolerated*
If the patient concurrently has ingested ethanol, then the ethanol loading dose must be modified so that the blood ethanol level does not exceed 100 to 150 mg/dL	

MAINTENANCE DOSE		
Patient	***Oral Ethanol 20%***	***Intravenous Ethanol 10%***
Non-drinker	*80 to 130 mg/kg/hr (0.5 - 0.8ml/kg/hr) PO or via nasogastric tube*	*80 to 130 mg/kg/hr (1.0 - 1.6 ml/kg/hr)*
Chronic alcoholic	*150 mg/kg/hr (0.9ml/kg/hr) PO or via nasogastric tube*	*150 mg/kg/hr (1.9ml/kg/hr)*
During dialysis	*250 to 350 mg/kg/hr (1.6 - 2.0ml/kg/hr) PO or via nasogastric tube*	*250 to 350 mg/kg/hr (3.1 - 4.4ml/kg/hr)*

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PREPARATION GUIDE FOR ORAL ETHANOL 20% V/V SOLUTION AND INTRAVENOUS ETHANOL 10% V/V SOLUTION (UNDENATURED)

	Undenatured alcohol 96% (for internal use only)
Indication	*Methanol poisoning*
Reconstitution	*Not required*
Further dilution	***Oral ethanol 20% v/v solution*** *125ml of alcohol 96% added with distilled water up to 600ml* ***IV ethanol 10% v/v solution*** *58ml of alcohol 96% added into 500ml of D5%*
Diluent	*Oral: distilled water, IV: D5%*
Storage & stability	*Store at room temperature, below 25°C* *After dilution:* *Oral: 3 months from the date of preparation or according to the expiry date of the raw material (whichever comes first)* *IV: 24 hours*

- *Treatment should be titrated to a blood ethanol level of 100mg/dL. If your hospital does not have blood ethanol measurement, titrate ethanol infusion to patient's clinical condition i.e. patient shows improvement without adverse effects.*
- *Patients must be monitored in ICU if IV ethanol infusion is given-potential adverse effects of hypotension, tachycardia, CNS/respiratory depression, hypoglycemia, electrolyte derangement, and gastritis.*
- *Ethanol infusion is preferably avoided in pediatric and pregnant patients.*

Dialysis :

- *Treatment with just ethanol/fomepizole infusion without hemodialysis has been successful in patients who are minimally toxic, without renal impairment. Dialysis is indicated if any of the following is present:*
  - *Severe metabolic acidosis (pH<7.20) which is persistent despite resuscitative measures and antidote administration*
  - *Signs of end organ toxicity oliguric or anuric renal failure*
  - *Blood methanol level of more than 50mg/dL*

Treatment of Suspected Cases :

- *Patients with symptoms of nausea, vomiting, drowsiness, inebriation, visual abnormalities and abdominal pain must be admitted for further investigation and work up to rule out other pathology.*
- *Blood and urine for methanol must be sent for all patients fulfilling criteria of suspected cases.*
- *Patients who are clinically well and asymptomatic may be discharged home with advice after notification.*

Prognosis:

- *Prognosis is generally good for patients who present early to hospital even in severe poisoning. Identification of the poisoning with rapid resuscitation, antidote administration and dialysis when indicated is crucial.*
- *Patients who present in a comatose state on arrival have been shown to have poor neurological recovery even with dialysis.(1)*
- *The presence of lesions in the putamen, thalamus, external capsule and white matter on CT brain are associated with poor clinical outcome.(2,3)*

Ref :

1. *Md Noor J, Hawari R, Mokhtar MF, Yussof SJ, Chew N, Norzan NA, et al. Methanol outbreak: a Malaysian tertiary hospital experience. International Journal of Emergency Medicine. 2020 Feb 7;13*
2. *Taheri MS, Moghaddam HH, Moharamzad Y, Dadgari S, Nahvi V. The value of brain CT findings in acute methanol toxicity. European Journal of Radiology. 2008 Dec 20;73(2):211–4*
3. *Sasani MR, Molavi Vardanjani H, Mehdipour Namdar Z, Jeddi M, Seif S, Sedighi S, et al. Prognosis of Methanol Poisoning in a Developing Setting. Archives of Iranian Medicine. 2024 Mar 1;27(3):127–34*